

LUCA MALAGUTTI

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🔗 [LucaMalagutti](https://github.com/LucaMalagutti)

Work Experience

Applied Scientist II

Oct. 2025 – Present

Thomson Reuters Labs

Zug, Switzerland

- Owned end-to-end technical roadmap for AI-powered news drafting system, productionizing AI features (article generation, urgent updates, spot stories) that processes thousands of news articles monthly.
- Led cross-functional technical integration across 4 AI workstreams, serving as primary technical contact between editorial and engineering teams.

Applied Scientist I

Oct. 2024 – Oct. 2025

- Built and deployed a scalable multi-source data ingestion pipeline for financial news alerting, leveraging LLMs to automatically transform unstructured company releases into news alerts at 30,000 per month scale.
- Implemented production monitoring and model evaluation infrastructure for LLM-based systems, ensuring reliability across multiple deployed features and establishing performance baselines.

Applied Science Intern

Apr. 2024 – Oct. 2024

- Developed and deployed a transformer-based news relevance classification model for financial alerting systems, achieving stakeholder-defined accuracy targets.
- Built LLM-as-Judge evaluation framework for RAG system quality assessment, establishing automated evaluation pipelines that reduced manual QA overhead.

Research Assistant

Oct. 2023 – Mar. 2024

ETH Zurich

Zurich, Switzerland

- Research Assistant supervised by Prof. Ryan Cotterell. Conducted research on tokenization, sampling methods, and language model generation, co-authoring papers at ICLR, ACL, EMNLP, and NAACL.

Data Science Intern

Mar. 2022 – Sep. 2022

SEI Novus

Zurich, Switzerland

- Built and deployed from scratch a complete NLP pipeline for automatic data extraction from financial documents.

Education

MSc in Computer Science

Sep. 2020 – Aug. 2023

ETH Zurich

Zurich, Switzerland

- Teaching Assistant for the Natural Language Processing & Large Language Models courses held at ETH Zurich. Delivered lectures and designed graded assignments and exam questions.
- Selected courses: Natural Language Processing, Advanced Machine Learning, Machine Perception, Big Data.

BSc in Computer Engineering

Sep. 2017 – Jul. 2020

Politecnico di Milano

Milan, Italy

- Final grade: 110/110 - Awarded Best Freshmen Award and merit-based tuition Scholarships.
- Selected courses: Probability and Statistics, Linear Algebra, Algorithms and Principles of Computer Science.

Selected Publications

- Juan Luis Gastaldi, John Terilla, **Luca Malagutti**, Brian DuSell, Tim Vieira, and Ryan Cotterell. "The Foundations of Tokenization: Statistical and Computational Concerns." *ICLR 2025*
- Luca Malagutti**, Andrius Buinovskij, Anej Svete, Clara Meister, Afra Amini, and Ryan Cotterell. "The Role of n -gram Smoothing in the Age of Neural Networks." *NAACL 2024*
- Clara Meister, Tiago Pimentel, **Luca Malagutti**, Ethan Wilcox, and Ryan Cotterell. "On the Efficacy of Sampling Adapters." *ACL 2023*

Skills & Interests

Programming Languages

ML & NLP

LLMs

MLOps

Languages

Endurance sports

Python, Java, C, SQL, Bash, L^AT_EX

PyTorch, HuggingFace Transformers, vLLM, spaCy, scikit-learn, pandas

Fine-Tuning, LoRA, Prompt Engineering, Evaluation, RAG, Agents

AWS, Docker, Git, CI/CD, Experiment Tracking, Data Pipeline Design

Italian (native), English (professional proficiency), German (B1 level)

Completed full Ironman, 3× Half Ironman, and 3 marathons (PB 2h49m)